Does the tangent cone $T(a, C)$ contain the point a?

Supplement to Chapter 4, Proposition 2.1

Definition 1 (Tangent cone, Def. 2.1)**.** Let $C \subset \mathbb{R}^n$ and $a \in C$. A vector $d \in \mathbb{R}^n$ belongs to $T(a, C)$ if there exist sequences $d_k \to d$ and $t_k \to 0^+$ such that $a + t_k d_k \in C$ for all $k \in \mathbb{N}$.

Question. Is it true that $a \in T(a, C)$ for every C and every $a \in C$?
Answer. No. The statement is false in general. Note that $T(a, C)$ is a set of *directions* (displacements measured from a), not of points of C; this is why it is natural for 0 to belong to it, while there is no reason for a itself to belong to it.

Proposition 1. *The statement "$a \in T(a, C)$ for all $C \subset \mathbb{R}^n$ and all $a \in C$" is false.*

Proof. It suffices to exhibit a set C and a point $a \in C$ with $a \notin T(a, C)$.

Let $C = \{x \in \mathbb{R}^2 : \ \|x\| \leq 1\}$ be the closed unit disk and $a = (0, 1) \in C$, so $\|a\| = 1$.

Step 1 (a necessary condition for tangent directions). Let $d \in T(a, C)$, with $d_k \to d$, $t_k \to 0^+$ and $a + t_k d_k \in C$. Then

$$1 \ \geq \ \|a + t_k d_k\|^2 = \|a\|^2 + 2t_k \langle a, d_k \rangle + t_k^2 \|d_k\|^2 = 1 + 2t_k \langle a, d_k \rangle + t_k^2 \|d_k\|^2.$$

Subtracting 1 and dividing by $t_k > 0$,

$$2\langle a, d_k \rangle + t_k \|d_k\|^2 \leq 0 \qquad \text{for all } k.$$

Since $d_k \to d$, the sequence $\|d_k\|^2$ is bounded, so $t_k \|d_k\|^2 \to 0$. By continuity of the inner product, letting $k \to \infty$ gives

$$\langle a, d \rangle \leq 0 \qquad \text{for every } d \in T(a, C).$$

Step 2 (contradiction). Suppose $a \in T(a, C)$. Taking $d = a$ in Step 1,

$$\langle a, a \rangle = \|a\|^2 = 1 \leq 0,$$

which is a contradiction. Hence $a \notin T(a, C)$. □

Remark (Consistency with Figure 2 of the notes). In fact $T(a, C) = \{d : \langle a, d \rangle \leq 0\} = \mathbb{R} \times (-\infty, 0]$ here, the half-plane shown in Figure 2. Since $a = (0, 1)$ has positive second coordinate, $a \notin \mathbb{R} \times (-\infty, 0]$. (Step 1 proves only the inclusion "$\subset$", which is all the counterexample needs.)

Remark (An even simpler counterexample). Let $C = \{a\}$ with $a \neq 0$. If $a + t_k d_k \in C$ then $t_k d_k = 0$, so $d_k = 0$ and $d = \lim d_k = 0$. Thus $T(a, C) = \{0\}$, and $a \notin T(a, C)$.

Remark (When it does hold). The inclusion $a \in T(a, C)$ is true in some special cases, for example:

- $a = 0$: then $a = 0 \in T(a, C)$ by Proposition 2.1.
- $a \in \operatorname{int} C$: there is $r > 0$ with $B(a, r) \subset C$; for any d, take $d_k = d$ and $t_k = \min\{1/k, \ r/(1 + \|d\|)\}$ to get $a + t_k d \in C$. Hence $T(a, C) = \mathbb{R}^n \ni a$.

So $a \in T(a, C)$ depends on C and a; it is not a property of the tangent cone.